

## Introduction

To guaranty that the provided text dataset is appropriately prepared, analyzed, modeled, and evaluated, a well-structured analysis strategy is essential. Building a model that can predict an author based on their writing style is the primary objective of this research. Quality data preparation, exploratory text analysis, appropriate text preprocessing, model training with balanced classes, and the appropriate assessment technique selection will be the main emphasis of the workflow. An LSTM-based classification neural network will be constructed and trained using TensorFlow, while the analysis portion will be handled using the Spark framework.

## Planned Workflow

|    | <i>STEP</i>                             | <i>PLANNED ACTIVITY</i>                                                     | <i>PURPOSE</i>                                                            |
|----|-----------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1  | <b>Dataset loading &amp; inspection</b> | Load with Spark; inspect row/column counts, schema and sample records.      | Confirm the file is parsed correctly and suitable for text modelling.     |
| 2  | <b>Data quality</b>                     | Check missing/blank values, duplicates and valid author labels.             | Prevent avoidable quality problems from entering the model.               |
| 3  | <b>EDA</b>                              | Analyse class distribution, sentence lengths and linguistic features.       | Understand patterns that may distinguish authors.                         |
| 4  | <b>Outliers</b>                         | Use IQR/boxplots and inspect extreme sentence lengths.                      | Decide whether extremes are errors or genuine style.                      |
| 5  | <b>Feature selection</b>                | Use Sentence as input and Author as target; exclude label-encoding columns. | Prevent target leakage and keep the task aligned to authorship from text. |
| 6  | <b>Balancing</b>                        | Randomly retain 10,000 sentences per author.                                | Create equal class representation and manageable training size.           |
| 7  | <b>Text cleaning</b>                    | Lowercase and remove punctuation/special characters.                        | Standardise the input text using the taught preprocessing approach.       |
| 8  | <b>Data split</b>                       | 80% train / 20% test, stratified by author.                                 | Keep a fully unseen test set while preserving class balance.              |
| 9  | <b>Tokenisation</b>                     | Fit a 5,000-word Keras tokenizer on training text only.                     | Convert words to integer sequences without test-set vocabulary leakage.   |
| 10 | <b>Padding</b>                          | Post-pad/truncate to 50 tokens.                                             | Create fixed-length LSTM inputs.                                          |
| 11 | <b>Baseline LSTM</b>                    | 64-unit LSTM + 5-class softmax; early stopping.                             | Establish a clear reference model.                                        |
| 12 | <b>Model improvement</b>                | Bidirectional LSTM + dropout + dense layer.                                 | Address baseline overfitting and improve generalisation.                  |
| 13 | <b>Final evaluation</b>                 | Accuracy, precision, recall, F1 and confusion matrix.                       | Measure overall and per-author test performance.                          |
| 14 | <b>Book Bot</b>                         | Apply identical preprocessing to new text and return probabilities.         | Demonstrate a usable final prediction pipeline.                           |
| 15 | <b>Report</b>                           | Summarise evidence, results, limitations and recommendations.               | Communicate the analysis clearly and professionally.                      |

## Planned Evaluation Metrics

Several categorization metrics will be used to assess the model. The percentage of sentences that are correctly categorized will be reported by accuracy. Recall will display the percentage of positive class phrases that the model correctly predicted, while precision will provide the ratio of projected positive classes that are truly positive.

Since the F1-score incorporates both prediction accuracy and precision, it will be the main statistic for model analysis. The model's overall performance and its capacity to categorize each author's phrases will be evaluated using both Macro F1 and Weighted F1 scores.

Additionally, the confusion matrix will be examined to show how similar the linguistic patterns of various writers are.

To determine if the model is overfitting or underfitting, the accuracy and loss from training and validation sets will be monitored for every epoch. Prior to evaluating the final model on the test set, this analysis will be completed.

### **Planned Model Improvement**

An conventional LSTM layer and output for each of the five author classes will be used to construct a baseline LSTM.

To determine if overfitting or underfitting is present, the baseline model's performance on the training and validation sets will be compared. If the model needs to be improved, a bidirectional LSTM will be constructed to allow the model to extract context from the words that come before and after it in a phrase. To reduce overfitting, the dropout regularization approach will be used.

The validation accuracy and loss will be used to compare the updated model with the baseline model. For the final test set, the model with the greatest performance will be chosen.